Vectors

I. Vector Basics

Defn: A **vector**, **v**, is a mathematical object that has both magnitude/length and direction.

Notation:

- Vectors will written either in bold or with an arrow: $\mathbf{v}$, $\vec{v}$.
- A vector, $\vec{v}$, in $\mathbf{R}^2$ ($\vec{v} \in \mathbf{R}^2$) can be represented by 2 coordinates: $\vec{v} = [v_1, v_2] \in \mathbf{R}^2$
- A vector, $\vec{v}$, in $\mathbf{R}^3$ ($\vec{v} \in \mathbf{R}^3$) can be represented by 3 coordinates: $\vec{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$
- The special unit vectors $\mathbf{i} = [1,0,0]$, $\mathbf{j} = [0,1,0]$, and $\mathbf{k} = [0,0,1]$ form a basis for $\mathbf{R}^3$, so we can write $\vec{v} = [v_1, v_2, v_3]$ as $\vec{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ (linear combination of **i**, **j**, **k**)

Ex:

- $[10,14], [0,1], [-1,2] \in \mathbf{R}^2$ and $[10,4] = 10\mathbf{i} + 4\mathbf{j}$
- $[1,1,-1], [0,0,0], [-10,12,-8] \in \mathbf{R}^3$ and $[-10,12,-8] = -10\mathbf{i} + 12\mathbf{j} - 8\mathbf{k}$

Thm: Let $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$ be vectors. Then

$$\mathbf{a} = \mathbf{b} \quad \textit{iff} \quad a_1 = b_1, a_2 = b_2, a_3 = b_3$$

Ex: $[2, 0, 2d + c] = [c, 0, 1] \implies c = 2$ and $2d + c = 1$, so $d = -\frac{1}{2}$

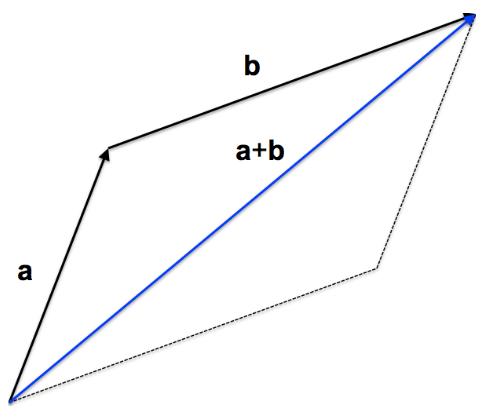
II. Vector Addition

Vector Operation 1: Addition

Let $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$

Then

$$\mathbf{a} + \mathbf{b} = [a_1, a_2, a_3] + [b_1, b_2, b_3] = [a_1 + b_1, a_2 + b_2, a_3 + b_3]$$



Note:

- $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3 \implies \mathbf{a} + \mathbf{b} \in \mathbf{R}^3$
- $\mathbf{a} + \mathbf{b}$ is known as the resultant of **a** and **b**. (e.g. If **a** and **b** were forces then $\mathbf{a} + \mathbf{b}$ would be the resultant force.

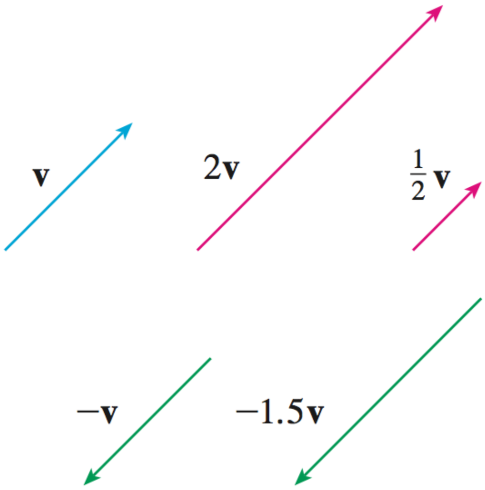
III. Scalar Multiplication

Property: Scalar Multiplication of Vectors

Let $\mathbf{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$ and $c \in \mathbf{R}$ (scalar)

Then

$$c\mathbf{v} = c[v_1, v_2, v_3] = [cv_1, cv_2, cv_3]$$



Note:

- Scalar multiplication can stretch , compress , or flip a vector.

IV. Length(Norm) of a Vector

Defn: Let $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2, w_3]$. Then the length of $\mathbf{v}$ and $\mathbf{w}$ (denoted $|\mathbf{v}|$ and $|\mathbf{w}|$) is given by

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2} \quad \text{and} \quad |\mathbf{w}| = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

Ex: Find the length of $\mathbf{w} = [1, -2, 4]$

$$|\mathbf{w}| = \sqrt{(1)^2 + (-2)^2 + (4)^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

Note:

- $\mathbf{w} \in \mathbf{R}^3$ (vector), but $|\mathbf{w}| \in \mathbf{R}$ (scalar)

Defn: $\mathbf{v}$ is a unit vector if $|\mathbf{v}| = 1$

Ex: Find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{w}$ above.

$$\mathbf{u} = \frac{\mathbf{w}}{|\mathbf{w}|} = \frac{1}{\sqrt{21}}[1, -2, 4] = \left[\frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}, \frac{4}{\sqrt{21}}\right]$$